10. EXPLORE ANTIVIRUS DETECTION TECHNIQUES

DATE: 01/04/2026

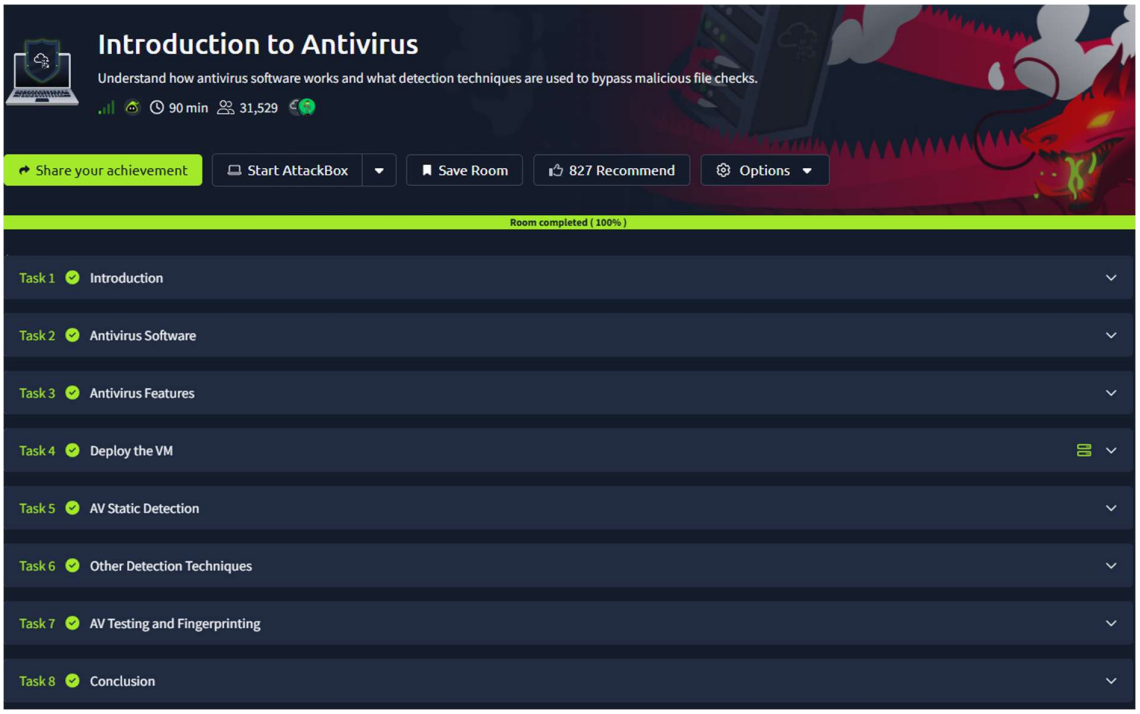
Aim:

To understand how antivirus software works and what detection techniques are used to bypass malicious file checks.

Procedure:

1. Log into TryHackMe and open the Introduction to Antivirus room.
2. Study the working principles of antivirus software.
3. Understand detection techniques such as signature-based and heuristic analysis.
4. Learn common methods used to bypass antivirus detection.
5. Analyze how malicious files are identified and handled.
6. Complete the tasks and answer the questions in the room.
7. Review the summary and confirm completion.

Tasks:



TASK 1 – Introduction

TASK 2 – Antivirus Software

Answer the questions below

What does AV mean?

Antivirus — ✓ Correct Answer

Which PC Antivirus vendor implemented the first AV software on the market?

McAfee — ✓ Correct Answer

Antivirus software is a _____-based security solution.

Host — ✓ Correct Answer

TASK 3 – Antivirus Features

Answer the questions below

Which AV feature analyzes malware in a safe and isolated environment?

Emulator — ✓ Correct Answer

An _____ feature is a process of restoring or decrypting the compressed executable files to the original.

unpacker — ✓ Correct Answer

Read the above to proceed to the next task, where we discuss the AV detection techniques.

No answer needed — ✓ Correct Answer

TASK 4 – Deploy The VM

Answer the questions below

Once you've deployed the VM, it will take a few minutes to boot up. Then, progress to the next task!

No answer needed — ✓ Correct Answer

TASK 5 – AV Static Detection

```
C:\Users\thm\Desktop\Samples>"c:\Program Files\ClamAV\sigtool.exe" --md5 AV-Check.exe
f4a974b0cf25dca7fbce8701b7ab3a88:6144:AV-Check.exe

C:\Users\thm\Desktop\Samples>_
```

```
C:\Users\thm\Desktop\Samples>strings AV-Check.exe | findstr "THM"
THM{Y0uC4nC-5tr16s}
```

Answer the questions below

What is the `sigtool` tool output to generate an MD5 of the `AV-Check.exe` binary?

f4a974b0cf25dca7fbce8701b7ab3a88:6144:AV-Check.exe ✓ Correct Answer

Use the strings tool to list all human-readable strings of the AV-Check binary. What is the flag?

THM{Y0uC4nC-5tr16s} ✓ Correct Answer

TASK 6 – Other Detection Techniques

Answer the questions below

Which detection method is used to analyze malicious software inside virtual environments?

Dynamic Detection ✓ Correct Answer

TASK 7 – AV Testing and Fingerprinting

Answer the questions below

For the C# AV fingerprint, try to rewrite the code in a different language, such as Python, and check whether VirusTotal flag it as malicious.

No answer needed ✓ Correct Answer

Read the Above!

No answer needed ✓ Correct Answer

TASK 8 – Conclusion

Answer the questions below

Congrats on completing the room, and keep learning!

No answer needed ✓ Correct Answer

Result:

Thus the TryHackMe room “Introduction to antivirus” was completed successfully.